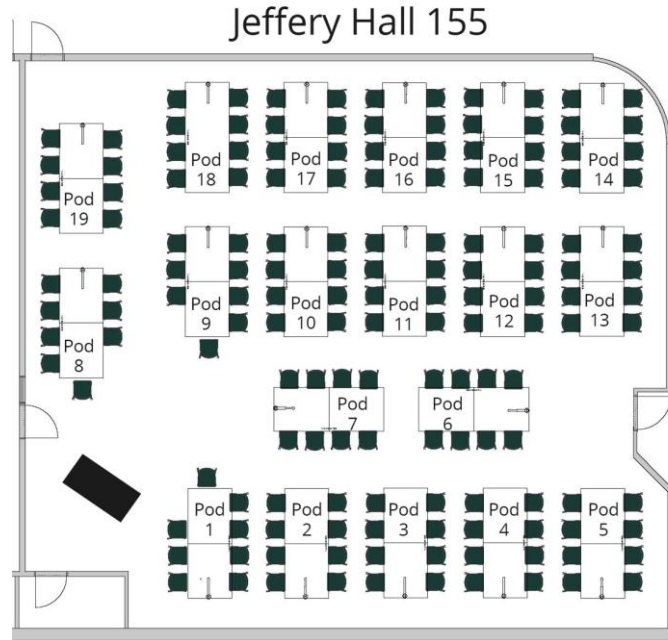


Engineering Design and Practice I: APSC-101

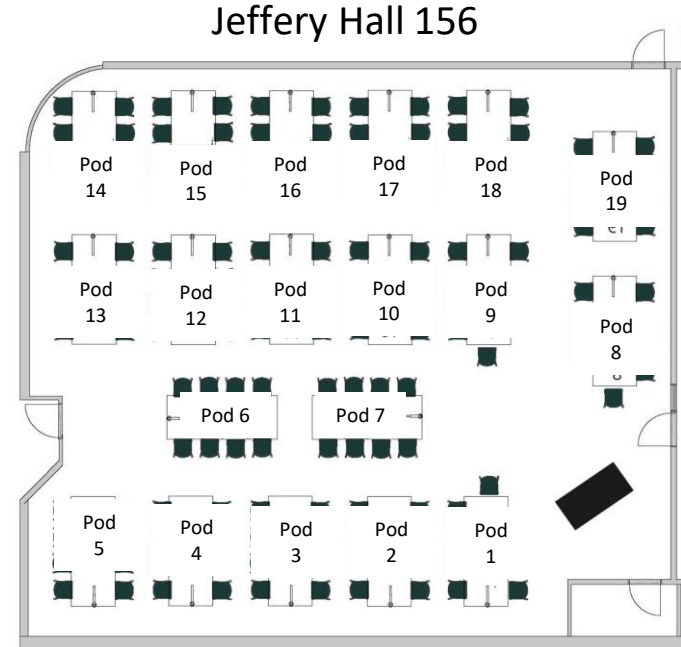
Welcome!

Please sit at your assigned table with your team. Find the email from Aphra that gives the room number (215/225/235) then your pod number (1A, 1B, etc.).

Then introduce yourselves to your teammates - share something about yourself!



Student Seats: 152



Student Seats: 152

We care about your safety!

- Exit routes, fire pull stations, and extinguisher locations
- Washroom locations
- Report any safety concerns to your instructors. Engineering Code of Ethics (Ontario): **“A practitioner shall regard the practitioner’s duty to public welfare as paramount.”**
- WHMIS training is a required element of this course
- In the event of an emergency, dial 911 or dial 613-533-6111 for Campus Security if on-campus. Campus Security are trained in First Aid, carry EpiPens and Naloxone. They will notify emergency services.
- If the building alarm sounds, take your belongings and leave by the nearest exit. Do not use elevators during evacuation. Move away from the building to prevent blocking the access for emergency responders. Never block aisles/stairs/doors with your personal belongings.

APSC 101: Engineering Design and Practice

Week 1

September 8-9, 2026

Dr. Paul Hungler, PEng



School

Electrical Engineering (BEng, RMC 2001)

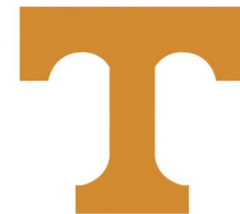
Chemical and Materials Engineering
(MAsc, RMC 2006)

Nuclear Engineering (PhD, RMC 2010)

Business Administration (MBA, UT, 2023)

Industry

Aerospace Engineering Officer (RCAF)



Dr. Paul Hungler, PEng



Research

Intelligent Dynamically Adaptive Simulation for Education and Training

Paul Hungler, PhD, PEng

Associate Professor

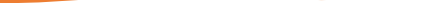
Department of Chemical Engineering and
Ingenuity Labs

Queen's University

Dupuis Rm 313

E-mail: paul.hungler@queensu.ca





Instructional team

Instructors



Brian Frank



Paul Hungler



Peter Gilbert

Course staff



Aphra Rogers



Michaela Olesova

Technical Staff



Vanessa Wood



Steve McNally

54 Project Managers

19 In-Studio mentors

14 Graders

2 Summer students

Avery and Vishwas

Lead Tech TA

Mohammedtufail Malik

19 Tech TAs

2 Teaming Coordinators

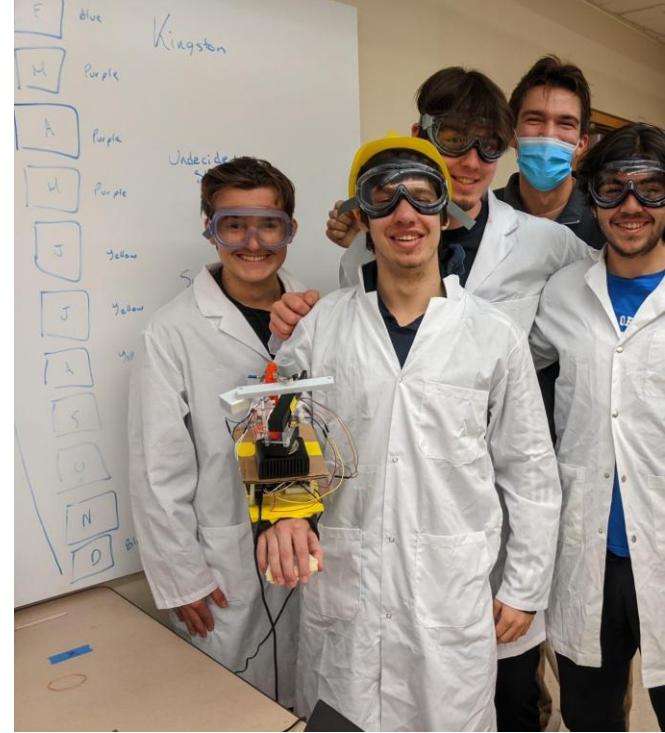
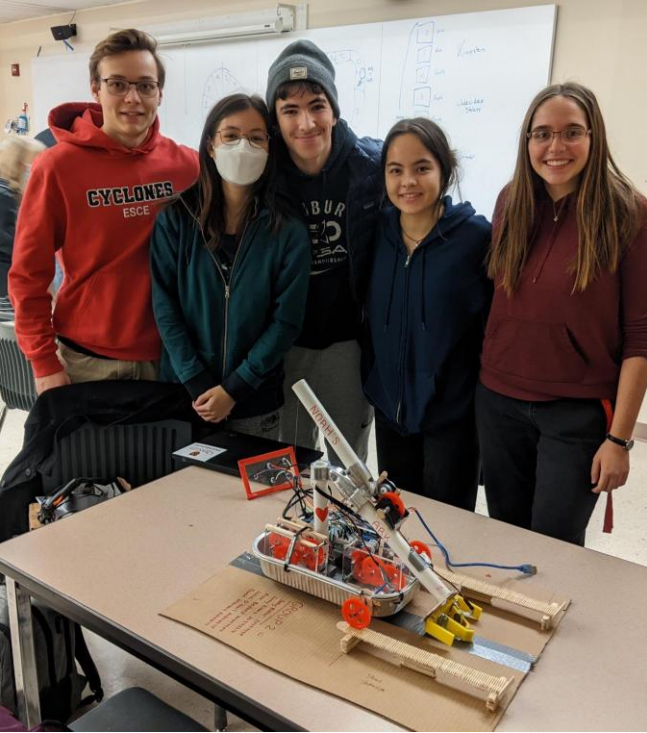
Zaid Kasim

Abdurrahman Sivesoglu

The background is a collage of three images. On the left, a woman in a dark lab coat stands in a laboratory, smiling. In the center, a robotic arm is positioned over a workbench. On the right, a person's hands are visible at a desk with a computer monitor displaying a 3D model.

Engineering: applying scientific principles and mathematical methods to solve real-world problems and create innovative solutions.

Often involves **Design, Analysis, Optimization, Inquiry,** and requires **collaboration, communication, and an awareness of social needs.**



COURSE PHILOSOPHY

Provide you situations that **emulate professional practice**,
and provide ***instruction, examples, and feedback*** to help
you solve **engineering problems**.

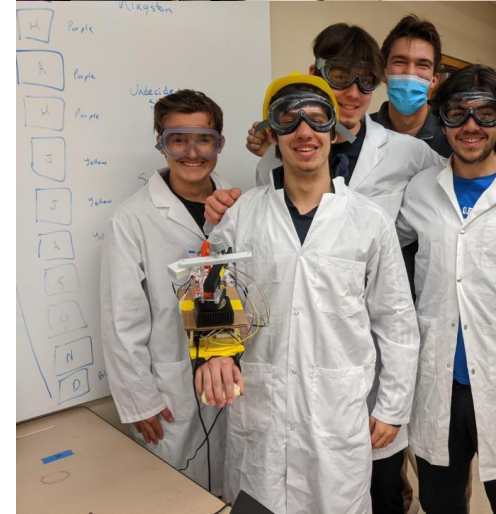
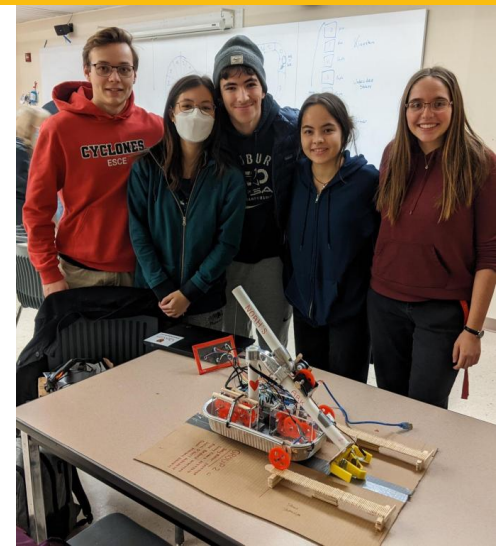
Your engineering experience

- How many have “reverse engineered” something to see how it works (electronics, code, appliance) (for school, club, work, or even on your own)?
- How many have worked to repair or improve something (electronics, code, appliance)?
- How many have tested something to evaluate its performance?
- How many have used math to model behaviour of some system?

Course approach

We present you a **challenging and motivating project**, provide **instruction, examples, activities**, and **mentorship in studios**, and support you with project managers outside class to develop the key competencies of

Engineering modeling
Engineering design
Collaboration
Professional behaviour
Professional growth

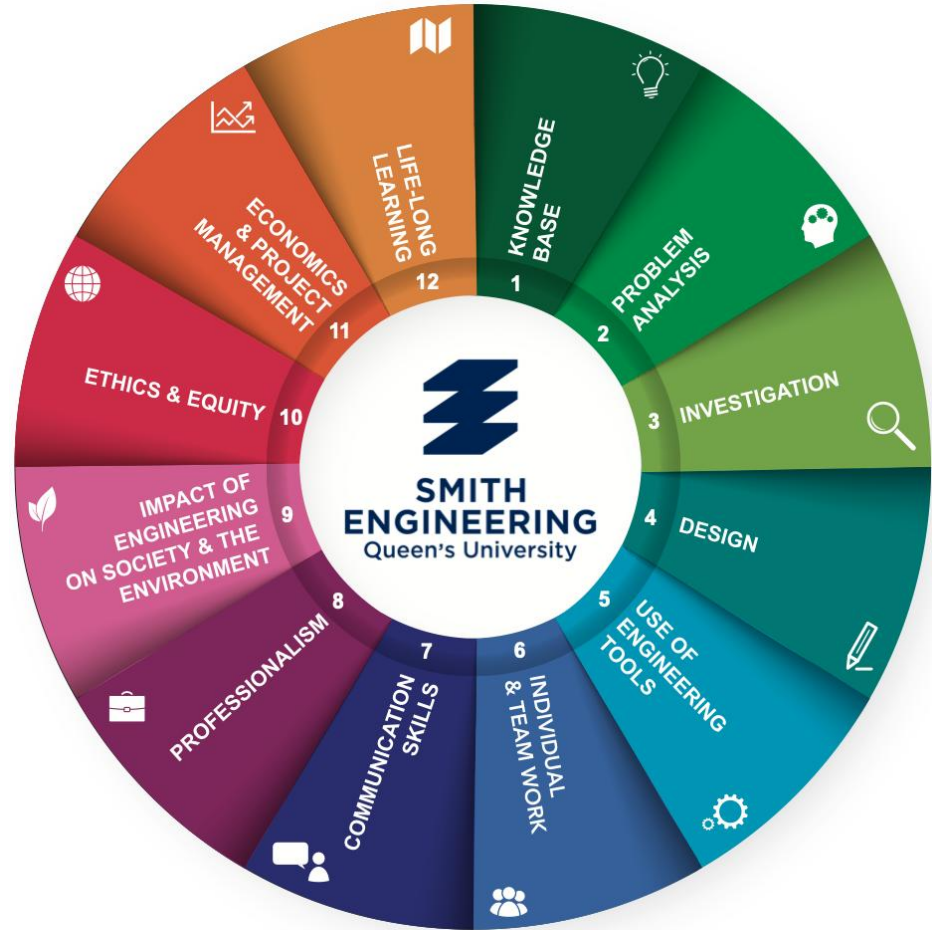


12 Graduate Attributes

A requirement for Accreditation is that graduates of Smith Engineering programs possess the 12 Graduate Attributes, which each have measurable Indicators linked to courses.

Most of your courses this term support Knowledge Base (#1) or Use of Engineering Tools (#5). This course addresses most other attributes; it is designed to be about “doing engineering”

The syllabus has specific references to the Indicators being tracked in this course. Please check them out. These form the basis of continual improvement in courses and programs.



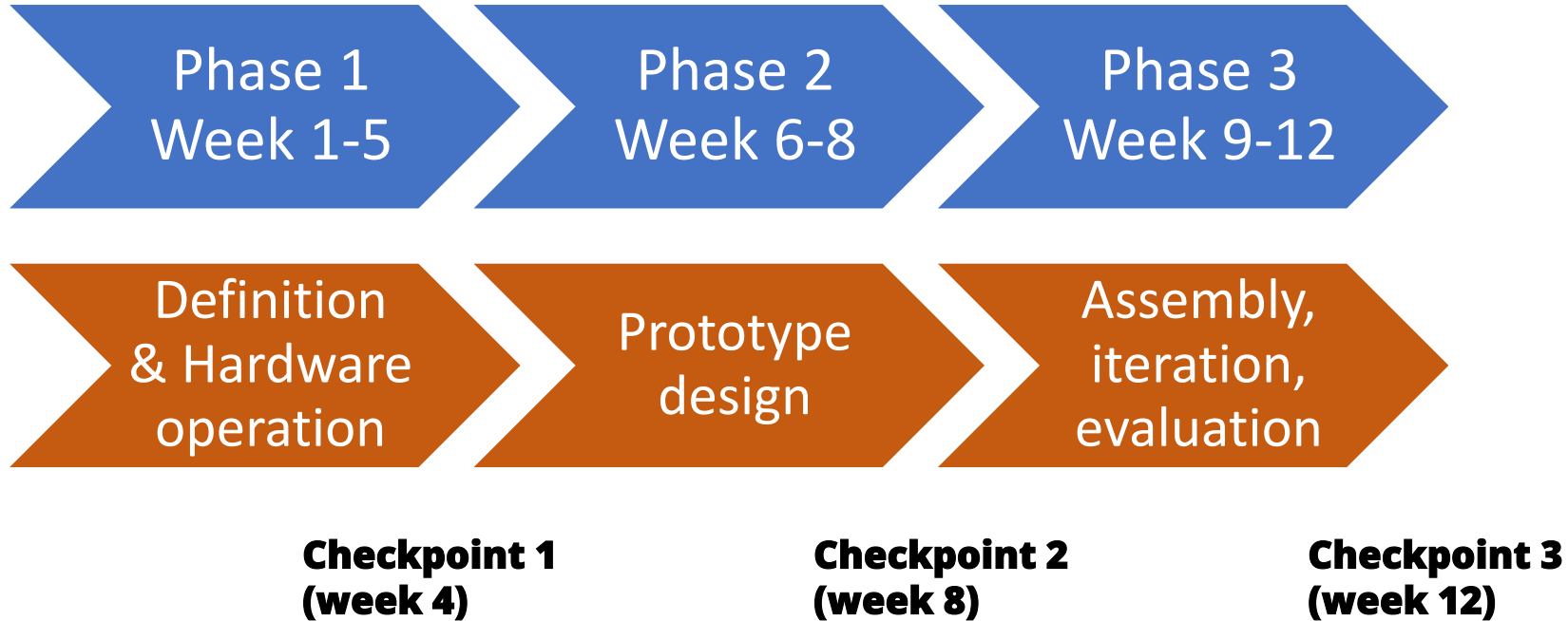
Course projects you selected in July

Drinking Water Filtration Prototype (~50% of teams): Design and implement a small prototype of a drinking water filtration and treatment plant tailored for communities lacking pre-existing municipal infrastructure, removing contaminant by flocculation and eliminating sediment from tanks.

Artificial Kidney Prototype (~25% of teams): Design and implement a prototype artificial kidney that will serve to regulate and filter contaminants within a blood substitute.

Remote Hydroponic Horticulture Prototype (~25% of teams): Design and build a scaled, automated hydroponic (soilless) gardening system prototype that monitors nutrient liquid, and automatically dispenses nutrients and balance pH of the solution.

Course timeline



Project Instructions

- Instructions common to all teams, and specific to each project, are provided
- **Safety and storage instructions are critical**

APSC 101: Engineering Design and Practice 1 | Fall 2026
Project Instructions



APSC 101 F2026 Design Projects

The goal of this term-length project is for you to work with your team to conceive, design, implement, test, and iterate a solution for a presented problem. To do this, you will develop skills in project planning, teamwork, idea generation, evaluating alternatives, and prototyping. Thanks to Avery Tribe and Vishwas Gowda who led the prototyping of these projects.

Project-Independent Instructions

The following instructions apply to all three design projects. Project-specific scenarios, performance requirements, individual subassembly roles, testing conditions, and optimization goals are provided in the project sections that follow.

Overall Project Goal

The goal of the term-length project is for your team to conceive, design, implement, test, and iterate a solution to an engineering problem. You will develop and demonstrate skills in problem framing, project planning, teamwork, idea generation, mathematical modeling, evaluating alternatives, CAD design and fabrication, prototyping, programming/control, testing, and evidence-based design improvement.

Individual Engineering Responsibility

Each team will divide primary responsibility among individual subassembly roles. Each role is an individual engineering responsibility within the integrated team design, and must provide meaningful opportunities for the student to make creative design decisions, create and iterate CAD-designed parts, and develop a mathematical model that informs the design. Each student will document their CAD work, model, design decision, and resulting evidence in their individual logbook and cite this evidence in the Individual Competency Portfolio.

Common Prototype Requirements

Your design kit contains pumps, motors, and sensors to design this system, and you have some prefabricated parts that you can use as you'd like; you will also design some of the subsystems yourself to be fabricated on a 3D printer. Note that we do not expect you can already do this – we will teach you what you need to know in the fall term! Requirements common to all projects include:

- Fit within a maximum footprint of 204 x 305 mm
- Complete all requirements in 10 minutes or less
- Include an Emergency Stop activated by a clearly visible and easily accessible button securely mounted on the prototype.
- Display the current operating stage and the relevant sensor measurements identified for the project.
- Be structurally rigid and mechanically assembled using removable methods such as friction fits or screws so that the system can be disassembled for reuse. Hot glue or other adhesives must not be used for the final prototype, but can be used in early iterations.
- Use 3D-printed material efficiently. Avoid unnecessary material and stress-concentrating features, and remain within the course limits on total printed material and printing time.
- Protect gears, moving components, and electronics with suitable enclosures or guards, while permitting maintenance and disassembly.
- Route and fasten wires and tubing so that they are protected, orderly, and do not interfere with

Syllabus

- Read the course syllabus for expectations, grading scheme, academic integrity, etc.

Welcome to Engineering Design and Practice 1 (APSC 101)! This course is structured around a design project that you will address over the semester with your team to develop confidence and skills in solving complex engineering problems, teaming, and managing projects. Your instructional team is excited to work with you!

TEACHING TEAM

Instructors		Staff	
Dr. Brian Frank, P.Eng. Course coordinator Office: WLH-701 brian.frank@queensu.ca Teams: frankb		Aphra Rogers, B. Sc., M.Sc. Program Lead Office: BMH-300 Tel: (613) 533-6000x77650 Apsc100.m1m3@queensu.ca	
Dr. Paul Hungler, P.Eng. Course Instructor Office: Dupuis Hall 313 Paul.hungler@queensu.ca		Michaela Olesova Office: BMH-300 Tel: (613) 533-6000x79517 Apsc100.m1m3@queensu.ca	
Dr. Peter Gilbert, P.Eng. Course Instructor Office: Dupuis Hall 425 p.gilbert@queensu.ca			

LAND ACKNOWLEDGEMENT

Queen's University is situated on traditional Anishinaabe and Haudenosaunee Territory.
See: <http://www.queensu.ca/encyclopedia/t/traditional-territories>

INCLUSIVITY STATEMENT

Queen's students, faculty, and staff come from every imaginable background – small towns and suburbs, urban high rises, Indigenous communities, and from more than 100 countries around the world. You belong here: <https://www.queensu.ca/inclusive/>.

Grading

Final Grade	Competency Review Outcome
F	By the Final Competency Review, does not demonstrate Meets Requirements in one or more of the five competencies.
D	By the Final Competency Review, demonstrates Meets Requirements in all five competencies. Some competencies may require additional coaching or reassessment before being achieved.
C	Demonstrates Meets Requirements in all five competencies throughout the Competency Review process. Evidence collected across the three checkpoints consistently supports competent first-year engineering practice.
B	Demonstrates Exceptional performance that is confirmed during the Competency Review process in at least two competencies. Exceptional performance is supported by multiple independent observations collected across the semester.
A-	Demonstrates Exceptional performance that is confirmed during the Competency Review process in at least three competencies.
A	Demonstrates Exceptional performance that is confirmed during the Competency Review process in at least four competencies. Evidence from multiple checkpoints consistently demonstrates mature engineering judgement and contributions that strengthen engineering outcomes.
A+	Demonstrates Exceptional performance that is confirmed during the Competency Review process in all five competencies . Evidence collected throughout the semester consistently demonstrates outstanding engineering capability, professionalism, collaboration, and continual improvement.

Standards for each competency posted in syllabus

Competency	Below Requirements	Meets Requirements	Exceptional
Engineering Modeling	Does not yet independently identify or apply appropriate engineering models...	Independently identifies appropriate engineering variables, assumptions, and quantitative relationships....	Demonstrates sophisticated engineering reasoning in assumptions and models, including ...
Engineering Design	Requires substantial guidance to define engineering problems...	Independently applies an appropriate engineering design process...	Demonstrates mature engineering judgement by anticipating risks, balancing competing stakeholder needs...
Professional Behaviour	Professional behaviours are inconsistent. Preparation, reliability, timely completion, safety, communication, or responsibility require regular prompting or intervention.	Consistently demonstrates professional behaviour expected of a beginning engineer...	Demonstrates professional leadership by consistently taking initiative, anticipating issues...
Collaborative Engineering	Contributions to the team are inconsistent or unreliable. Communication, participation, or responsiveness limits overall team effectiveness.	Consistently contributes to team goals through reliable participation, respectful communication, ...	Consistently strengthens team effectiveness by facilitating productive collaboration...
Professional Growth	Self-assessment is incomplete, weakly supported by evidence, or improvement plans are vague...	Accurately evaluates personal engineering performance using evidence, identifies meaningful priorities for improvement...	Demonstrates sophisticated professional self-regulation by accurately calibrating strengths and limitations...

A group of colorful wooden human figures standing in a line, representing a diverse team. The figures are in various colors including blue, yellow, red, green, and brown. They are standing on a light-colored surface against a blurred background.

Team formation and peer
feedback activities

Today: starting team formation

One of the course competencies we'll be developing is **Collaboration**. You will be evaluating yourself and your team members on six dimensions at each checkpoint to support **Being a Team PLAYER**

- **Performance:** Contribute equitably to the team's work, uphold high standards and develop new capabilities.
- **Leadership:** Know the team's purpose, fulfill your formal/informal roles, and help the team stay focused.
- **Attitude:** Show enthusiasm in the team's work and foster a collaborative atmosphere.
- **HarmonyY:** Promote constructive conflict and diffuse interpersonal conflict.
- **Expression:** Communicate effectively and share ideas and information.
- **Reliable:** Demonstrate reliability and integrity.

We will be starting on this today.



Forming inclusive teams

- Beware of stereotypical roles
- Value diversity in all forms on your team (gender, orientation, language, ethnicity, ...)
- Be supportive of those whose English skills are still developing who will have to work harder in the course.
- Be aware that cultural or religious practices that may interfere with group work.

Overall: Thoughtful and considerate

Activity 1: Team Shared Strength and Risks (10 minutes)

- Each person creates a logbook entry for studio 1, with date. Records key points of discussion in your logbook.
- Each person introduces themselves and briefly describes three things (no discussion, interruption):
 - Contribution “One thing this team can rely on me for is...” (technical skill or process strength)
 - Support need “One thing I may need help with is...”
 - Work preference: “I do my best work when...”
- Team Synthesis: As a group, identify
 - Shared Strength: “As a team, we are strong at...”
 - Shared risk: “A likely challenge for us is...”
 - One Working Norm (one actionable rule) “To address this, we will...”
- **Examples:** Shared strength: “We have strong measurement and organization skills.” Shared risk: “We may rush decisions without checking data.” Working norm: “We will pause before conclusions and have the skeptic confirm evidence.”

TAs and instructor will circulate and introduce themselves throughout this activity.

If you complete the activity and have already chatted with a TA or instructor, begin reviewing your Project instructions in OnQ (Content > Project Instructions)

Primary assignments you'll be submitting

1. Team Engineering Design Portfolio

- Checkpoint 1 (end of week 4)
- *Checkpoint 2 (end of week 8)*
- Checkpoint 3 (end of week 12)

We will give feedback and teams will iterate on:

1. Problem definition, needs, safety
2. Team charter
3. Engineering Requirements & Decisions
4. *Engineering Design*
5. *Validation, Plan* and Results
6. Improvement

APSC 101: Engineering Design and Practice 1 | Fall 2026



Team Engineering Design Portfolio

Throughout the term your team will maintain and update a living engineering design portfolio that shows the thought process in completing design activity. This document should not contain every discussion or idea by the team. Rather it is intended to both support the team in design and explain the decisions made by the team. This document provides guidance on what to include in this portfolio. You are free to make a copy of this document to use as a template for your portfolio.

Cover Page

This should include

- Project title
- Team members
- Client
- Project scenario
- Current checkpoint
- Submission date

Revision History

This should include a table with headings like below, which is updated each time it is submitted for review. If you are asked to resubmit the document for revision after a checkpoint, ensure that is recorded.

Version	Date	Checkpoint	Summary of Changes
---------	------	------------	--------------------

Primary student deliverables

2. Individual Competency Portfolio

At each checkpoint each person will submit their portfolio:

1. Evaluate current performance in each course competency
2. Identify specific evidence supporting their self-evaluation;
3. Explain why that evidence demonstrates the competency;
4. Identify most important areas for development; and
5. Establish specific actions to take next

APSC 101: Engineering Design and Practice 1 | Fall 2026



Individual Competency Portfolio (ICP)

The Individual Competency Portfolio (ICP) is your opportunity to evaluate your current development as an engineer using evidence from your work. At each checkpoint you will provide a personal evaluation of your performance on each competency and plan for further development. A description of each checkpoint is below.

Checkpoint 1: Concept Review

At this checkpoint, you will:

1. evaluate your current performance in each course competency;
2. identify specific evidence supporting your evaluation;
3. explain why that evidence demonstrates the competency;
4. identify your most important areas for development; and
5. establish specific actions you will take before Checkpoint 2.

This submission is the primary artifact used to evaluate your individual competency at this checkpoint, using evidence from your work. Your evaluation should describe your performance to this point, considering the pattern of your work and behaviour to date, including evidence for future improvement needs. Evidence to describe may include specific pages or entries in your individual engineering logbook, specific sections, requirements, models, or decisions in your team's Engineering Design Portfolio, specific actions during team meetings or studios, feedback from your studio mentor or project manager, peer evaluations, completion of quizzes and safety requirements and overall role with the team. Rather than stating that you "worked on" something, explain **what you did, what it demonstrates, and where the evidence can be found.**

Physical Logbook

- Bring it to all class sessions, team meetings, and working sessions to record actions and ideas.
- It shows what engineering work you personally did, how your thinking and design developed, and quantitative or experimental evidence informing decisions
- Requirements include sequentially numbered pages, contain dated, chronological entries, maintained regularly as engineering work occurs, legible and organized so that an evaluator can locate and understand evidence you reference
- You will submit it at each checkpoint with your Individual Competency Portfolio; it is a key source of evidence of your contributions

APSC 101: Engineering Design and Practice 1 | Fall 2026
Physical Logbook Instructions



Physical Logbook instructions

Your Individual Engineering Logbook is a chronological working record of your own engineering activity and thinking throughout the project. It serves two purposes:

1. to help you develop and maintain a professional record of your engineering work; and
2. to provide evidence of your individual contribution and competency within a team project, particularly when you submit your Individual Competency Portfolio (ICP) at each checkpoint. At each competency checkpoint, you will be asked to reference specific logbook page numbers that provide evidence for your competency claims.

The logbook is a working engineering document, in which you record your work as you do it. It should show how your understanding, analysis and design develop over time—including ideas that do not work as expected.

Ensure you bring it to all class sessions, team meetings, and working sessions to record actions and ideas.

Your logbook should show clearly show:

- what engineering work you personally did
- how your thinking and design developed
- quantitative or experimental evidence informing your decisions
- what you learned and what changed as the project progressed

Logbook requirements

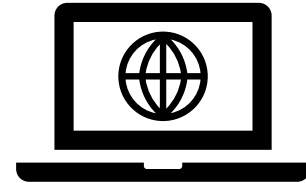
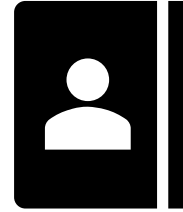
Your logbook must:

- have sequentially numbered pages;
- contain dated, chronological entries;
- clearly identify work that you personally performed;
- be maintained regularly as engineering work occurs rather than reconstructed before a checkpoint;
- preserve mistakes, unsuccessful approaches and changes in thinking rather than replacing them with a cleaned-up final account; and
- be sufficiently legible and organized that an evaluator can locate and understand evidence you reference.

Your logbook does **not** need to be polished or resemble a formal report. Rough calculations, annotated sketches, diagrams and concise notes are appropriate.

Bring to class (101/199)

- Design logbook and laptop (with charger if necessary) each week
- Bring adapter to connect to HDMI cable if necessary (note: you'll need to be able to plug a USB-A cable to your Arduino too).
- Some times in class will be device free.



Professionalism and Academic Integrity

You are expected to use the utmost respect in your dealings with your colleagues and instructors, and when participating in activities, discussions, and online communication.

Queen's University values maintaining an environment free of, and will not tolerate, harassment, discrimination, and reprisal. The Student Code of Conduct applies to all students at Queen's, and this is one of the required competencies in the course.

Engineers have a duty to

Act at all times with devotion to the high ideals
of personal honour and professional integrity
Give proper credit for engineering work

*Professional Engineers Ontario Code of Ethics,
Section 77 of the O. Reg. 941*

The material presented in this course is the property of the instructor. Copying this material for distribution (e.g. uploading material to a commercial third-party website) can lead to a violation of Copyright law.

Departures from Academic Integrity (DFAI)

DFAI includes plagiarism, use of unauthorized materials or services, facilitation, forgery, falsification, unauthorized use of intellectual property, collaborating during individual assessments, sharing material with unauthorized parties, including online tutoring services, etc. (This is not an exhaustive list).

Academic Integrity in APSC 101/199

Specific requirements are in the course syllabus.

Brief version:

CHAT WITH OTHERS FOR IDEAS.

DO YOUR OWN WORK BY YOURSELF (OR WITH TEAM).

DON'T COPY SOMEONE ELSE'S (this is “facilitation” in DFAI policy).

DON'T SHARE YOUR FILES WITH SOMEONE ELSE.

The work in a submission should have been done by the people whose names are on the document, aside from properly cited material.

The penalty for departing from academic integrity is a zero on the activity, and a grade point deduction in your course grade.

Any use of generative AI (e.g. ChatGPT) needs to be described in submissions unless it's an activity we do in class (e.g. we will be using GenAI as part of design activity).

Accommodations/Considerations

Follow provided instructions from Faculty office for accommodations/considerations.

Note that in this course, 7-day accommodations are generally not possible as many assessments are team-based, or scaffold into the next week's assessment.

Wellness and Mental Health Resources for Students

See also the [EngWell Hub](#)

Eng Resources	Queen's Resources	External Resources
<u>Wellness Advisor</u> Unsure of the type of support and resources you need? Our Wellness Advisor can determine your individual needs and connect you with the appropriate resources and support.	<u>Queen's Student Wellness Services</u> Medical, mental health, accessibility and health lifestyle appointments. Phone: 613-533-2506 or Visit: in person 1st Floor Mitchell Hall	<u>Empower Me</u> Empower Me provides Queen's students with virtual access to therapy, coaching and 24/7 crisis support. Access through the Dialogue App
<u>Academic Accommodation and Consideration Advisor (absences)</u> Understand and navigate the policies and procedures for Requests for Academic Consideration (RAC) and Requests for Excused Absence for Significant Events (REASE).	<u>Peer Support Centre</u> The Peer Support Centre offers drop-in hours for students to speak with supportive and compassionate student volunteers.	<u>GOOD2TALK</u> text GOOD2TALKON to 686868 to chat with a trained volunteer Crisis Responder. The service is free, confidential and available 24/7.
<u>Embedded Counsellor & BIPoC Counsellor</u> Are you looking for short-term counselling to work on personal issues? Book in with one of our Embedded Counsellors.	<u>Faith and Spiritual Life</u> Offers multi-faith, non-judgmental secular and non-secular support for spiritual, personal, and financial problems, concerns or crises.	<u>Headspace</u> Headspace is the most science-based meditation app available, and it can assist you with your work-life balance goals, including managing stress, anxiety, getting better sleep, and more.
<u>Neurodivergent Student Supports</u>		

For Student resources and Smith Engineering Policies and Procedures (applying to all courses), please refer to the [EngQ Hub on onQ](#) or the [Teaching & Student Support Hub](#)



Managing constructive feedback

Being able to give and receive feedback is important when working with others, as a student and an engineer.

This activity will help you improve your skills for

- giving feedback to your teammates without excessive emotional reactions
- receiving feedback from others without your emotions preventing you from being able to “hear” it
- constructing and delivering actionable feedback



For feedback to be useful, it should:



Not be an offload of emotion: Develop awareness of your emotions - do you feel angry, frustrated, disrespected? What made you feel this way? How might others perceive the situation?



Be specific: Provide observations or examples, not generalisations.



Occur at the right time: Feedback should be given promptly unless strong emotions or other constraints make a delay more appropriate.



Be usable: State how the person can implement your feedback, connecting to the objectives or strategies of the team / task.



Be desired: Check when and how your peers want feedback to build an environment where feedback most effective and seen as an opportunity for growth.

Handout and plastic bag

- 1 page per person
- 1 bag per team – have one person pickup



HANDOUT A Preparing and Giving Feedback

●●●—3T PLAY

(Interview your PAIR before the group task) What skills or strengths does your pair expect to contribute to the group task?

1. ...

2. ...

What do they expect to find difficult during the group task?

1. ...

2. ...

Generating feedback for your pair (after the group task)

What were your pair's strengths and contributions during the group task? Cite specific instances of how/when your pair contributed or made you feel that your contributions were appreciated. Include at least one really positive point.

•

•

How could your pair have improved their contributions or better supported your participation? Cite specific instances and be sure to find at least one really critical or negative point to help them improve.

•

•

For feedback to be useful and constructive, it should:



Not be an offload of emotion: Develop awareness of your emotions - do you feel angry, frustrated, disrespected? What made you feel this way? How might others perceive the situation?



Be specific: Provide observations or examples, not generalisations.



Occur at the right time: Feedback should be given promptly but strong emotions or other constraints may make a delay more appropriate.



Be usable: State what the person can do to implement your feedback, connecting to the objectives or strategies of the team / task.



Be desired: Check when and how your peers want feedback to build an environment where feedback most effective and seen as an opportunity for growth.

Activity Outline

HANDOUT A Preparing and Giving Feedback

(Interview your PAIR before the group task) What skills or strengths does your pair expect to contribute to the group task?

1. ...

2. ...

What do they expect to find difficult during the group task?

1. ...

2. ...

1. On your team, form 1 pair and 1 triad (three people). On triads, person 1 observes person 2 who observes person 3 who observes person 1.
2. Do the preparation activity on the top of HANDOUT A with your pair/triad, based on planning for the following activity:



Activity time!

1. With your all teammates, use the materials in the bag to build a spaghetti marble track
 - 15 minutes to complete your build
 - You can add support materials (e.g. waterbottles, mugs, etc.) but the marble can only touch pasta and playdough, and must always be in contact with the track
 - Competition for the fastest marble track of 1.5m (length of string)
 - You can catch the marble at the end of the track



2 minutes to test

- Test your track
- If you have a successful run, write your team number and time on the whiteboard at the front of the room



Plan your feedback

1. Working alone, write down feedback for your pair on HANDOUT A. Be sure to apply the guidelines for useful feedback!



We all feel emotions during feedback



FIGHT

Breathe.

Use active listening
before reacting.

Try to understand the
other's point of view.

Avoid "Yes but..."



FLEE

Breathe.

Remind yourself
feedback is an
opportunity to learn.

Ask for clarification +
future oriented advice.



FREEZE

Breathe.

Move physically, head,
hands or feet.

Notice your emotions.
Ask for time to reflect.



Doing feedback

1. Review what you have written on “Handout A” to align with good practices for constructing feedback.
2. In turns, share your feedback with your pair. Be sure to include at least one very positive point and one quite critical point. Remain attentive to your own emotions and your pairs.

Analyze what you experienced

1. Complete “Handout B” to reflect on your experience.

HANDOUT B Receiving Feedback and Reflecting

What feedback did you receive from your pair? Note specific points, including at least 1 positive aspect and one negative aspect.

-
-

What were your emotions when RECEIVING feedback?

-
-

What emotion management techniques did you use?

-
-

What were your emotions when GIVING feedback?

-
-

What emotion management techniques did you use?

-
-

How does this experience compare to other times you have given or received feedback from teammates? What aspects of this experience could you use to improve the feedback culture in your next team ?

-
-

Fear of feedback – managing emotions



FIGHT

Breathe.

Use active listening before reacting.

Try to understand the other's point of view.

Avoid "Yes but..."



FLEE

Breathe.

Remind yourself feedback is an opportunity to learn.

Ask for clarification + future oriented advice.



FREEZE

Breathe.

Move physically, head, eyes, hands or feet.

Notice your emotions.

Ask for time to reflect.

Please clean up materials

Scour the floor and tables for any leftover bits!

Course schedule

Phase	Weeks	Actions	Deliverables
1	1-4	Define the problem including needs and requirements, and show control of mechatronic system	OnQ quizzes Checkpoint 1 Team and Individual Portfolios
	5	Design requirements	3D Print run #1: Calibration and test part
2*	6	Intense design activity iteratively redefines the problem and create conceptual ideas for subsystems	OnQ quizzes
	7	The intense design activity iteratively redefines the solution; CAD support in APSC 162 lab timeslots	OnQ quizzes Design concept in CAD studio
	8	Design iteration and finalization; CAD support in APSC 162 lab timeslots.	Show proof of concept in CAD studio 3D Print run #2: Critical Functionality Checkpoint 2 Team and Individual Portfolios
3	9-10	Supportive design, iteration, and professional practice activities	OnQ quizzes Design scenario test (week 9) Incremental checkpoint reviews mentors & PMs 3D Print run #3 (week 10): Repair and structural elements
	11	Assembly, comparison to requirements, and iteration	
	12	Showcase demonstration! (In APSC 162): Working Drawings & Complete Project drawings of assemblies and full assembly, including dimensioning and tolerances	Showcase demonstration! Checkpoint 3 Team and Individual Portfolios

*Weeks 6, 7, 8 use some timeslots from APSC 162

Teaming space: Microsoft Teams

As part of this week's activities, you'll create a shared team cloud space - suggest Microsoft Teams in your Queen's Office365 account.

Instructions posted in Week 1 material in OnQ.

Apps



Outlook



OneDrive



Word



Excel



PowerPoint



OneNote



SharePoint



Teams



Class Notebook



Sway

Team meeting template

In OnQ > Content > Project Instructions

APSC 101 Team Meeting Template

This template is to record activities and decisions in your team meetings. It should be submitted as an appendix to your Team Engineering Design Portfolio as evidence of team activities. **Teams should use these pages as a template for their project manager and weekly team meetings, to be filled in before and during the meeting.**

Week 1 Meeting with Project Manager (use this as a template for meeting with project manager)

Date: Click or tap to enter a date.

Team #:

Location:

Project Manager:

Start Time:

End Time:

Team Meeting Attendance:

Action Items from meeting

Indicate the action items arising from the meeting

Action Item	
1.	
2.	
3.	
4.	
5.	

Project Hardware!!



Safety is critical –follow safety protocol
Follow instructions for parts (e.g. pH sensor needs to be stored wet)

APSC101 2026 Parts List ~ Hydroponics Project

Loose in box:

- Arduino Mega Board
- USB cable A-B
- Motor shield
- Wall adapter power supply
- Breadboard
- OLED Display
- Submersible pump
- Peristaltic pump
- Diaphragm Pump
- Servo motor
- TDS Sensor
- pH Sensor
- Liquid level sensor
- Acrylic Base
- Acrylic
- Container 500ml
- Container 300ml
- Container Square
- Velcro stickers (10)
- Syringe
- Syringe plug
- Tubing 3x5mm (~1m)
- Tubing 6x8 mm (30cm)
- Jumpers M-M 12" (4)
- Jumpers M-M 8" (10)
- Jumpers F-M 12" (4)
- Jumpers F-M 8" (10)

Clear Clam Shell

- 5"×5" nylon cloth
- DC motor
- Red and black spade connectors
- Gears – Variety Pack
- Worm gear
- Funnel
- Barb connectors (2)
- Filter sponge
- Yarn (for motor curves)
- Spool (for motor curves)
- Marrette connector (2)
- Splice Connector (2)

Chemical Supplies (Hand out Week 6)

- pH buffer solution
- Baking Soda
- Miracle-Gro

Bag 1: Electronics

- 2-pin push button (2)

Bag 2: Hardware

- Standoffs Small (6)
- Magnets (4)
- 4-40 mm screws (10)
- M3x16 mm screws (10)
- 4-40 Nuts (10)
- M3 Nuts (10)
- Dowel shaft
- 12 zip ties (2 large, 10 small)

There is a safety guide for each project: read and follow it

WATER TREATMENT SAFETY GUIDE

Student quick guide | Read the supplied SDSs and follow course instructions

MATERIALS

Bentonite clay

RISK	Dust irritates eyes/airways; silica may be present.
FIRST AID	Eyes: rinse 15 min. Inhalation: fresh air; seek help if symptoms persist.
DISPOSE	Seal and return unused clay and collected spills.

Aluminum sulfate (alum)

RISK	Acidic; may seriously damage or irritate eyes.
FIRST AID	Eyes: rinse 15 min. Skin: wash.
DISPOSE	Return unused alum in labelled bottle; no drain.

Bentonite and alum flocc slurry

RISK	Wet clay is slippery; avoid eye and skin contact.
FIRST AID	Eyes/skin: rinse 15 min; seek help if needed.
DISPOSE	Return slurry in the labelled liquid-waste bottle.

Used filter and filter cake

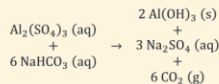
RISK	Wet solids are slippery; dry residue creates dust.
FIRST AID	Eyes: rinse 15 min. Skin: wash. Inhalation: fresh air.
DISPOSE	Seal wet solids in supplied bag; return.

PROCESS CONTROLS

SAFE WORKFLOW

- Spill tray; keep away from food.
- Wear goggles, nitrile gloves, closed shoes.
- Add whole clay dose gently.
- Dispose of waste solutions in return waste bottle.
- Rinse equipment with water; DO NOT USE ANY CLEANING PRODUCTS
- Return waste bottles.
- Return unused solids in original bottles.

EXPECTED REACTIONS



- $\text{Al}(\text{OH})_3$ floc captures bentonite.
- Sulfate remains dissolved; pH falls.
- No dangerous heat or gas is expected.

SPILLS AND FIRST AID

- Stop pumps before cleanup.
- Dampen dry clay; collect without sweeping.
- Keep all solids out of drains.
- Poison Centre 1-800-268-9017; call 911 in an emergency;
- Campus emergency response: 613-533-6111

RETURN LIQUID AND SOLIDS SEPARATELY. NO SINKS, DRAINS, OR GARBAGE.

HYDROPONIC MODEL SAFETY GUIDE

Student quick guide | Read the supplied SDSs and follow course instructions

MATERIALS

Miracle-Gro granules

RISK	Eye/skin irritant; harmful if swallowed.
FIRST AID	Eyes/skin: rinse 15 min. Inhaled: fresh air. Swallowed: call Poison Centre or a doctor.
DISPOSE	Return unused material in original bottle; return used material in waste bottle

0.05 M sodium bicarbonate solution

RISK	Eye irritant; acidic mixtures may release carbon dioxide.
FIRST AID	Eyes: rinse 15 min; seek care if pain persists. Skin: wash with soap and water.
DISPOSE	Return unused material in the original bottle.

Certipur pH 4.01 sachet

RISK	Potassium hydrogen phthalate solution; non-hazardous.
FIRST AID	Eyes/skin: rinse 15 min. Inhaled: fresh air. Swallowed: drink water; seek care if unwell.
DISPOSE	Return used/unused buffer in the waste bottle.

Certipur pH 7.00 sachet

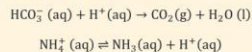
RISK	Dilute phosphate solution; non-hazardous.
FIRST AID	Eyes/skin: rinse 15 min. Inhaled: fresh air. Swallowed: drink water; seek care if unwell.
DISPOSE	Return used/unused buffer in the waste bottle.

PROCESS CONTROLS

SAFE WORKFLOW

- Wear goggles, nitrile gloves, closed shoes.
- Use spill tray; keep away from food.
- Target pH 5.8-6.2; stop at 6.5.
- Dispose of waste solutions in return waste bottle.
- Dispose of buffer solutions in separate waste bottle.
- Rinse equipment with water; DO NOT USE ANY CLEANING PRODUCTS
- Return waste bottles.

EXPECTED REACTIONS



- Acid plus bicarbonate releases CO_2 .
- $\text{pH} > 8$ will produce significant amount of ammonia and precipitate nutrients
- Never use bleach or seal vessels.

SPILLS AND FIRST AID

- Stop pumps before cleanup.
- Absorb liquid; sweep solids without raising dust.
- Poison Centre 1-800-268-9017; call 911 in an emergency.
- Campus emergency response: 613-533-6111

RETURN ALL WASTE TO COURSE. NO SINKS, DRAINS, GARBAGE, OR RECYCLING.

Risk is based on both probability of incident and *consequence* –we mitigate by minimizing at least one.

Risk	Mitigation strategy
Overheating/fire from short circuit	Do not short circuit wires from wall adapter Check wiring before powering on Keep water below and away from electrical currents Don't use electrical components near highly flammable material
Injury from mechanical motion	Remove loose jewelry, tie long hair back Prevent access to moving parts
Injury from tool use, material	Read product instructions, safety information, learn proper use of tools and materials Access to data sheets (MSDS)

Your team will need to meet to plan team activities

BMH has Group Rooms you can book for meetings. Your Project Manager can help with guidance



[Home](#) > [About](#) > [Facilities And Labs](#) > [Group Room Booking](#)

Use [the room booking system to reserve a group room](#)

Group rooms are working offices for student teams. They are simply furnished, with a table, chairs, whiteboard and network connections. There are 38 group rooms; most accommodate 6 students; some will seat 10 or 12. [Our online booking system](#), is available for groups to reserve the rooms up to several weeks ahead of time.

Group rooms may be booked by students for a wide range of activities, such as brainstorming, design discussions, preparation and presentation of reports, meetings with clients, and meetings with industry representatives and anyone else visiting the ILC. Outside the academic year they become workspaces for projects and outreach programs.



Reminders

This week:

- First meeting with your project manager this week. Plan your team's time to meet about doing the hardware assembly work (you can book group rooms in BMH)
- Read the syllabus
- Review the documents in OnQ under Content > Project Instructions to know what you're working toward over the next 5 weeks (including the project instructions document, Engineering Team Design Portfolio, and Individual Competency Portfolio)

For next week:

- APSC 199 (Engineering Communications) class on Monday- bring your 1st presentation
- Hardware technical support drop in sessions are available, see calendar in onQ.

Upcoming major deliverables:

- Checkpoint 1: Engineering Team Design Portfolio and Individual Competency Portfolio

Next APSC 101 class is week 4. Keep checking your onQ announcements!